Vector 3D: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Vector basics

Formulas

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|}, \qquad |x\hat{i} + y\hat{j} + z\hat{k}| = \sqrt{x^2 + y^2 + z^2}$$

Vector of length k along $\vec{a}$: $k\hat{a}$

KEY POINTS

- Vector = magnitude + direction; $\vec{AB}$: tail A, head B; line of support = line AB
- Null vector $\vec{0}$: zero length, any direction; unit vector: $|\hat{a}| = 1$
- Equal vectors: same length and direction; in $\hat{i}, \hat{j}, \hat{k}$ form compare components
- Angle between vectors: bring tails together; $0 \le \theta \le \pi$
- Collinear: $\vec{a} = \lambda\vec{b}$, i.e. $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$; like if $\lambda > 0$, unlike if $\lambda < 0$
- Vector parallel to $\vec{p}$: take $\lambda\vec{p}$
- Coplanar: lie in one plane; any two vectors are coplanar; free vectors may be shifted parallel

2. Addition and scalar multiple

Formulas

$$\big||\vec{a}| - |\vec{b}|\big| \le |\vec{a} + \vec{b}| \le |\vec{a}| + |\vec{b}|$$

Closed polygon: $\vec{AB} + \vec{BC} + \vec{CA} = \vec{0}$

KEY POINTS

- Parallelogram on $\vec{a}, \vec{b}$: diagonals $\vec{a} + \vec{b}$ and $\vec{b} - \vec{a}$
- Add component-wise; additive inverse of $\vec{a}$ is $-\vec{a}$
- Right equality: $\vec{a}, \vec{b}$ like parallel; left equality: unlike parallel
- $|\lambda\vec{a}| = |\lambda||\vec{a}|$; direction kept if $\lambda > 0$, reversed if $\lambda < 0$

3. Coordinates in 3D

Formulas

Section: $\vec{r} = \frac{m\vec{b} + n\vec{a}}{m + n}$; centroid $\frac{\vec{a} + \vec{b} + \vec{c}}{3}$

$$|AB| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

KEY POINTS

- $P(x, y, z)$ has position vector $x\hat{i} + y\hat{j} + z\hat{k}$; $\vec{AB} = \vec{b} - \vec{a}$; 8 octants by signs
- On x-axis: $y = z = 0$; in xy-plane: $z = 0$ (similarly others)
- Distance of P from x-axis $\sqrt{y^2 + z^2}$; from xy-plane $|z|$
- Foot on xy-plane $(x, y, 0)$; foot on x-axis $(x, 0, 0)$

4. Linear dependence in a plane

Formulas

$\vec{r} = x\vec{a} + y\vec{b}$ (unique) for $\vec{a}, \vec{b}$ non-collinear

KEY POINTS

- Dependent: $x_1\vec{a}_1 + \cdots + x_n\vec{a}_n = \vec{0}$ with some $x_i \neq 0$; else independent
- Two vectors dependent $\Leftrightarrow$ collinear; three $\Leftrightarrow$ coplanar; any four in space are dependent
- $x\vec{a} + y\vec{b} = x'\vec{a} + y'\vec{b}$ with $\vec{a}, \vec{b}$ independent $\Rightarrow x = x',\ y = y'$
- Points $\vec{a}, \vec{b}, \vec{c}$ collinear $\Leftrightarrow x\vec{a} + y\vec{b} + z\vec{c} = \vec{0}$, $x + y + z = 0$, not all zero

5. Vector equation of a line

Formulas

$$\vec{r} = \vec{a} + t\vec{b}, \qquad \vec{r} = \vec{a} + t(\vec{b} - \vec{a}), \qquad \vec{r} \times \vec{b} = \vec{a} \times \vec{b}$$

Bisectors of $\angle(\vec{a}, \vec{b})$: internal $\lambda(\hat{a} + \hat{b})$, external $\lambda(\hat{a} - \hat{b})$

KEY POINTS

- $\vec{a}$ = a point on the line, $\vec{b}$ = direction; general point $\vec{a} + t\vec{b}$
- Two lines: equate general points; consistent $t, s \Rightarrow$ intersecting, else skew (if not parallel)
- $\vec{b}_1 \parallel \vec{b}_2$: parallel or coincident (check a common point)
- Bisector of angle A in $\triangle ABC$: $\vec{r} = \vec{a} + \lambda \left(\dfrac{\vec{AB}}{|\vec{AB}|} + \dfrac{\vec{AC}}{|\vec{AC}|} \right)$

Watch out

$\hat{a} + \hat{b}$ (unit vectors), not $\vec{a} + \vec{b}$, gives the bisector

6. Dot product and projection

Formulas

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Projection of $\vec{a}$ on $\vec{b}$: $\dfrac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$; vector component $\dfrac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2}\vec{b}$

KEY POINTS

- $\hat{i} \cdot \hat{i} = 1$, $\hat{i} \cdot \hat{j} = 0$; commutative, distributive; $\vec{a} \cdot \vec{a} = |\vec{a}|^2$
- $\vec{a} \perp \vec{b} \Leftrightarrow \vec{a} \cdot \vec{b} = 0$; acute $\Leftrightarrow > 0$; obtuse $\Leftrightarrow < 0$ (not parallel)
- $|\vec{a} \pm \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 \pm 2\vec{a} \cdot \vec{b}$
- $\vec{a} + \vec{b} + \vec{c} = \vec{0} \Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{1}{2}\left(|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2\right)$
- Component of $\vec{a}$ perpendicular to $\vec{b}$: $\vec{a} - \dfrac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2}\vec{b}$
- Projection of segment AB on a line with DCs l, m, n: $|l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)|$

7. Cross product and area

Formulas

$$\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin\theta\, \hat{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2 |\vec{b}|^2$$

KEY POINTS

- $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$; $\vec{b} \times \vec{a} = -\vec{a} \times \vec{b}$
- $\vec{a} \times \vec{b} = \vec{0} \Leftrightarrow \vec{a} \parallel \vec{b}$ (or one is $\vec{0}$)
- Unit vectors perpendicular to both: $\pm \dfrac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$
- Area: parallelogram $|\vec{a} \times \vec{b}|$; triangle $\frac{1}{2}|\vec{AB} \times \vec{AC}|$; quadrilateral $\frac{1}{2}|\vec{d_1} \times \vec{d_2}|$
- $\vec{a} + \vec{b} + \vec{c} = \vec{0} \Rightarrow \vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$

Watch out

Cross product is not associative; $\vec{a} \times \vec{b} = \vec{a} \times \vec{c}$ does not give $\vec{b} = \vec{c}$

8. Direction cosines and ratios

Formulas

$$l^2+m^2+n^2=1, \qquad \cos^2\alpha+\cos^2\beta+\cos^2\gamma=1, \qquad \sin^2\alpha+\sin^2\beta+\sin^2\gamma=2$$

$$\cos\theta=\frac{|a_1a_2+b_1b_2+c_1c_2|}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$$

KEY POINTS

- DCs from DRs a,b,c: $\pm\dfrac{(a,b,c)}{\sqrt{a^2+b^2+c^2}}$; DRs through A,B: $(x_2-x_1,y_2-y_1,z_2-z_1)$
- Parallel: $\dfrac{a_1}{a_2}=\dfrac{b_1}{b_2}=\dfrac{c_1}{c_2}$; perpendicular: $a_1a_2+b_1b_2+c_1c_2=0$
- Equal angles with the axes: DCs $\pm\frac{1}{\sqrt{3}}(1,1,1)$
- Two linear and quadratic relations in l,m,n: eliminate one, solve the ratio

9. Line in Cartesian form

Formulas

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}=t, \qquad \frac{x-x_1}{x_2-x_1}=\frac{y-y_1}{y_2-y_1}=\frac{z-z_1}{z_2-z_1}$$

Distance of P from $\vec{r}=\vec{a}+t\vec{b}$: $\dfrac{|(\vec{p}-\vec{a})\times\vec{b}|}{|\vec{b}|}$

KEY POINTS

- Line through a point perpendicular to $\vec{b}_1,\vec{b}_2$: direction $\vec{b}_1\times\vec{b}_2$
- Points at distance r on the line: $(x_1+lr,\ y_1+mr,\ z_1+nr)$ with DCs l,m,n

STEPS

1. Foot of perpendicular: take $Q(x_1+at,\ y_1+bt,\ z_1+ct)$
2. Put $\vec{PQ}\cdot(a,b,c)=0$, find t
3. Image $P'=2Q-P$; distance $=PQ$

Watch out

A zero denominator means that coordinate is constant, not division by zero

10. Two lines: intersect, skew, shortest distance

Formulas

$$d=\frac{|(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)|}{|\vec{b}_1\times\vec{b}_2|}$$

Parallel lines: $d=\dfrac{|(\vec{a}_2-\vec{a}_1)\times\vec{b}|}{|\vec{b}|}$

KEY POINTS

- Intersect (coplanar) $\Leftrightarrow\begin{vmatrix}x_2-x_1 & y_2-y_1 & z_2-z_1\\ a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\end{vmatrix}=0$, lines not parallel
- Skew: neither parallel nor intersecting $\Leftrightarrow d>0$ and $\vec{b}_1\times\vec{b}_2\neq\vec{0}$
- SD line is along $\vec{b}_1\times\vec{b}_2$; its feet: $\vec{PQ}\cdot\vec{b}_1=\vec{PQ}\cdot\vec{b}_2=0$ with general points
- Point of intersection: equate general points, solve two equations, check the third

11. Equation of a plane

Formulas

$$\vec{r}\cdot\hat{n}=d, \qquad (\vec{r}-\vec{a})\cdot\vec{n}=0, \qquad ax+by+cz+d=0, \qquad \frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

Through three points: $\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_2-x_1 & y_2-y_1 & z_2-z_1 \\ x_3-x_1 & y_3-y_1 & z_3-z_1 \end{vmatrix} = 0$

KEY POINTS

- (a,b,c) = DRs of the normal; in $\vec{r}\cdot\hat{n}=d$, $d\geq 0$ is the distance from O
- Through $\vec{a}$ parallel to $\vec{b},\vec{c}$: $\vec{r}=\vec{a}+\lambda\vec{b}+\mu\vec{c}$, i.e. $[\vec{r}-\vec{a}\ \ \vec{b}\ \ \vec{c}]=0$
- Through A,B perpendicular to a plane with normal $\vec{n}$: normal $\vec{AB}\times\vec{n}$
- Parallel to yz-plane: $x=k$; containing x-axis: $by+cz=0$; parallel to x-axis: $by+cz+d=0$
- Intercepts a,b,c: area of the triangle cut $\frac{1}{2}\sqrt{a^2b^2+b^2c^2+c^2a^2}$

12. Angles with planes

Formulas

Two planes: $\cos\theta = \dfrac{|\vec{n}_1\cdot\vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$

Line and plane: $\sin\theta = \dfrac{|\vec{b}\cdot\vec{n}|}{|\vec{b}||\vec{n}|}$

KEY POINTS

- Planes parallel: $\dfrac{a_1}{a_2}=\dfrac{b_1}{b_2}=\dfrac{c_1}{c_2}$; perpendicular: $a_1a_2+b_1b_2+c_1c_2=0$
- Line parallel to plane: $\vec{b}\cdot\vec{n}=0$; line perpendicular to plane: $\vec{b}\parallel\vec{n}$
- Line lies in plane: $\vec{b}\cdot\vec{n}=0$ and one point of the line satisfies the plane

Watch out

Line-plane angle uses sin (normal makes $90^\circ-\theta$)

13. Distance, foot, image, bisector planes

Formulas

$$p=\frac{|ax_1+by_1+cz_1+d|}{\sqrt{a^2+b^2+c^2}},\qquad d_{\parallel}=\frac{|d_1-d_2|}{\sqrt{a^2+b^2+c^2}}$$

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}=-k\,\frac{ax_1+by_1+cz_1+d}{a^2+b^2+c^2}$$

$k=1$: foot; $k=2$: image

$$\frac{a_1x+b_1y+c_1z+d_1}{\sqrt{a_1^2+b_1^2+c_1^2}}=\pm\frac{a_2x+b_2y+c_2z+d_2}{\sqrt{a_2^2+b_2^2+c_2^2}}$$

KEY POINTS

- Parallel-plane distance: make a,b,c identical first
- P,Q same side $\Leftrightarrow$ $ax+by+cz+d$ has the same sign at both
- With $d_1,d_2>0$: the $+$ sign gives the bisector containing the origin
- Then $a_1a_2+b_1b_2+c_1c_2>0$: origin bisector is obtuse; <0: acute
- Line meets plane: put the general point of the line in the plane, solve for t
- Distance from plane measured along a line: go along that line to the plane, then measure

14. Family of planes, line of intersection, images

Formulas

Through $P_1=0$, $P_2=0$: $P_1+\lambda P_2=0$; line direction $\vec{n}_1\times\vec{n}_2$
Line of greatest slope: direction $(\vec{n}_H\times\vec{n}_G)\times\vec{n}_G$

KEY POINTS

- Unsymmetrical to symmetric form: direction $\vec{n}_1\times\vec{n}_2$, a point by putting $z=0$ (or $x=0$)
- Plane containing line $(\vec{a},\vec{b})$ and point $\vec{p}$: normal $(\vec{p}-\vec{a})\times\vec{b}$

- Two intersecting lines, or a line parallel to another: normal $\vec{b}_1 \times \vec{b}_2$; two parallel lines: $(\vec{a}_2 - \vec{a}_1) \times \vec{b}$
- Image of a line in a plane: meeting point and image of one more point (parallel line: image of one point, same direction)
- Projection of a line on a plane: use feet instead of images; image of a line in a line: images of two points
- Image of plane P_1 in P_2: $P_1 + \lambda P_2 = 0$ through the image of a point of P_1 (parallel: equal distance on the other side)
- Greatest slope line lies in the given plane G and is perpendicular to its meet with the horizontal plane H

Watch out

$P_1 + \lambda P_2 = 0$ misses P_2 itself; check it separately

15. Scalar triple product, volume, tetrahedron

Formulas

$$[\vec{a}\ \vec{b}\ \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Volume: parallelepiped $|[\vec{a}\ \vec{b}\ \vec{c}]|$; tetrahedron $\frac{1}{6}|[\vec{AB}\ \vec{AC}\ \vec{AD}]|$

KEY POINTS

- Cyclic: $[\vec{a}\ \vec{b}\ \vec{c}] = [\vec{b}\ \vec{c}\ \vec{a}]$; swapping two changes sign; dot and cross interchange
- Two equal (or parallel) vectors $\Rightarrow$ STP $= 0$; $[\lambda\vec{a}\ \vec{b}\ \vec{c}] = \lambda[\vec{a}\ \vec{b}\ \vec{c}]$
- $[\vec{a}+\vec{b}\ \ \vec{b}+\vec{c}\ \ \vec{c}+\vec{a}] = 2[\vec{a}\ \vec{b}\ \vec{c}]$; $[\vec{a}-\vec{b}\ \ \vec{b}-\vec{c}\ \ \vec{c}-\vec{a}] = 0$
- $[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}] = [\vec{a}\ \vec{b}\ \vec{c}]^2$; $[\vec{a}\ \vec{b}\ \vec{c}]^2$ = Gram determinant of dot products
- $[\vec{a}\ \vec{b}\ \vec{c}][\vec{p}\ \vec{q}\ \vec{r}] = \det(\text{dot products } \vec{a}\cdot\vec{p}, \dots)$
- Coplanar $\vec{a}, \vec{b}, \vec{c}$ (dependent) $\Leftrightarrow [\vec{a}\ \vec{b}\ \vec{c}] = 0$; four points: $[\vec{AB}\ \vec{AC}\ \vec{AD}] = 0$
- Tetrahedron: centroid $\dfrac{\vec{a}+\vec{b}+\vec{c}+\vec{d}}{4}$; height = projection of an edge on the face normal
- Regular tetrahedron, edge a: $V = \dfrac{a^3}{6\sqrt{2}}$, height $a\sqrt{\frac{2}{3}}$, angle between faces $\cos^{-1}\frac{1}{3}$, opposite edges perpendicular
- $\vec{AB}\cdot\vec{CD} + \vec{AC}\cdot\vec{DB} + \vec{AD}\cdot\vec{BC} = 0$: two pairs of opposite edges perpendicular $\Rightarrow$ third pair too

16. Vector triple product and four vectors

Formulas

$$\vec{a}\times(\vec{b}\times\vec{c}) = (\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c}, \qquad (\vec{a}\times\vec{b})\times\vec{c} = (\vec{a}\cdot\vec{c})\vec{b} - (\vec{b}\cdot\vec{c})\vec{a}$$

$$(\vec{a}\times\vec{b})\cdot(\vec{c}\times\vec{d}) = (\vec{a}\cdot\vec{c})(\vec{b}\cdot\vec{d}) - (\vec{a}\cdot\vec{d})(\vec{b}\cdot\vec{c})$$

$$(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d}) = [\vec{a}\ \vec{b}\ \vec{d}]\vec{c} - [\vec{a}\ \vec{b}\ \vec{c}]\vec{d} = [\vec{a}\ \vec{c}\ \vec{d}]\vec{b} - [\vec{b}\ \vec{c}\ \vec{d}]\vec{a}$$

KEY POINTS

- $\vec{a}\times(\vec{b}\times\vec{c})$ lies in the plane of $\vec{b}, \vec{c}$ and is $\perp \vec{a}$
- VTP is not associative: equal only if $\vec{a}, \vec{c}$ parallel or $\vec{b} \perp \vec{a}, \vec{c}$
- Vector in plane of $\vec{p}, \vec{q}$ and $\perp \vec{r}$: $\lambda\,\vec{r}\times(\vec{p}\times\vec{q})$
- $\hat{i}\times(\vec{a}\times\hat{i}) + \hat{j}\times(\vec{a}\times\hat{j}) + \hat{k}\times(\vec{a}\times\hat{k}) = 2\vec{a}$
- $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d})$ lies along the meet of the planes of $\vec{a}, \vec{b}$ and $\vec{c}, \vec{d}$

17. Linear combination in space and vector equations

Formulas

$$\vec{r} = x\vec{a} + y\vec{b} + z\vec{c}, \quad x = \frac{[\vec{r}\ \vec{b}\ \vec{c}]}{[\vec{a}\ \vec{b}\ \vec{c}]},\ y = \frac{[\vec{a}\ \vec{r}\ \vec{c}]}{[\vec{a}\ \vec{b}\ \vec{c}]},\ z = \frac{[\vec{a}\ \vec{b}\ \vec{r}]}{[\vec{a}\ \vec{b}\ \vec{c}]}$$

Reciprocal: $\vec{a}' = \dfrac{\vec{b}\times\vec{c}}{[\vec{a}\ \vec{b}\ \vec{c}]}$ (cyclic); $\vec{r} = (\vec{r}\cdot\vec{a}')\vec{a} + (\vec{r}\cdot\vec{b}')\vec{b} + (\vec{r}\cdot\vec{c}')\vec{c}$

KEY POINTS

- Needs $\vec{a}, \vec{b}, \vec{c}$ non-coplanar; coefficients are unique
- $\vec{a}\cdot\vec{a}' = 1$, $\vec{a}\cdot\vec{b}' = 0$; $[\vec{a}'\ \vec{b}'\ \vec{c}'] = \dfrac{1}{[\vec{a}\ \vec{b}\ \vec{c}]}$
- Extremum: $|\vec{a}-\vec{b}|^2 + |\vec{b}-\vec{c}|^2 + |\vec{c}-\vec{a}|^2 = 3\sum|\vec{a}|^2 - |\vec{a}+\vec{b}+\vec{c}|^2$ (≤ 9 for unit vectors)

STEPS

1. Isolate the unknown; dot with a vector to kill a cross term, cross to bring in a VTP
2. $\vec{r}\times\vec{a} = \vec{b}$ ($\vec{a}\cdot\vec{b} = 0$): $\vec{r} = \dfrac{\vec{a}\times\vec{b}}{|\vec{a}|^2} + \lambda\vec{a}$
3. $\vec{r}\cdot\vec{a} = p$, $\vec{r}\times\vec{b} = \vec{c}$: $\vec{r} = \dfrac{\vec{a}\times\vec{c} + p\,\vec{b}}{\vec{a}\cdot\vec{b}}$
4. $\vec{x} + \vec{a}\times\vec{x} = \vec{b}$: $\vec{x} = \dfrac{\vec{b} + (\vec{a}\cdot\vec{b})\vec{a} - \vec{a}\times\vec{b}}{1+|\vec{a}|^2}$
5. Else write $\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a}\times\vec{b})$ and compare

18. Common mistakes

STEPS

1. Using DRs as DCs without dividing by $\sqrt{a^2+b^2+c^2}$
2. Taking cos for the line-plane angle (it is sin)
3. Distance between parallel planes without equalising a, b, c
4. Skew-line SD formula used for parallel lines ($\vec{b}_1\times\vec{b}_2 = \vec{0}$)
5. Sign slip in $\vec{a}\times(\vec{b}\times\vec{c})$ vs $(\vec{a}\times\vec{b})\times\vec{c}$
6. Forgetting $\frac{1}{6}$ for a tetrahedron, $\frac{1}{2}$ for a triangle, and the modulus for volume
7. Bisector-plane rule applied before making both constants d_1, d_2 positive
8. Equating ratios with a zero denominator, or dropping $\pm$ in unit vectors
9. Two lines with non-parallel directions assumed to intersect (check STP)